

nRF9161-DEV-BOARD | nRF9161-LACA-R7 | Hierarchical schematic

the 1990s, the number of people in the world who are under 15 years of age has increased by 150 million. The number of people aged 15 and over has increased by 1.2 billion. The number of people aged 65 and over has increased by 100 million. The number of people aged 75 and over has increased by 50 million. The number of people aged 85 and over has increased by 20 million. The number of people aged 95 and over has increased by 10 million. The number of people aged 100 and over has increased by 5 million. The number of people aged 105 and over has increased by 2 million. The number of people aged 110 and over has increased by 1 million. The number of people aged 115 and over has increased by 500,000. The number of people aged 120 and over has increased by 250,000. The number of people aged 125 and over has increased by 125,000. The number of people aged 130 and over has increased by 62,500. The number of people aged 135 and over has increased by 31,250. The number of people aged 140 and over has increased by 15,625. The number of people aged 145 and over has increased by 7,812.5. The number of people aged 150 and over has increased by 3,906.25. The number of people aged 155 and over has increased by 1,953.125. The number of people aged 160 and over has increased by 976.5625. The number of people aged 165 and over has increased by 488.28125. The number of people aged 170 and over has increased by 244.140625. The number of people aged 175 and over has increased by 122.0703125. The number of people aged 180 and over has increased by 61.03515625. The number of people aged 185 and over has increased by 30.517578125. The number of people aged 190 and over has increased by 15.2587890625. The number of people aged 195 and over has increased by 7.62939453125. The number of people aged 200 and over has increased by 3.814697265625. The number of people aged 205 and over has increased by 1.9073486328125. The number of people aged 210 and over has increased by 0.95367431640625. The number of people aged 215 and over has increased by 0.476837158203125. The number of people aged 220 and over has increased by 0.2384185791015625. The number of people aged 225 and over has increased by 0.11920928955078125. The number of people aged 230 and over has increased by 0.059604644775390625. The number of people aged 235 and over has increased by 0.0298023223876953125. The number of people aged 240 and over has increased by 0.01490116119384765625. The number of people aged 245 and over has increased by 0.007450580596923828125. The number of people aged 250 and over has increased by 0.0037252902984619140625. The number of people aged 255 and over has increased by 0.00186264514923095703125. The number of people aged 260 and over has increased by 0.000931322574615478515625. The number of people aged 265 and over has increased by 0.0004656612873077392578125. 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D. GNSS

Trial	Control	MCI	AD
1	85	75	65
2	82	72	62
3	78	68	58
4	76	66	56
5	75	65	55

J. GPIO EXPANSION

Circumstance	Percentage of Respondents (%)
If someone is attacking you	45
If someone is threatening you	55
If someone is using force against you	65
If someone is using a weapon against you	75
If someone is using a gun against you	85

E. SIM/UICC

Circumstance	Justified (%)	Not justified (%)
If someone is attacking you	85	15
If someone is threatening you	75	25
If someone is harassing you	65	35
If someone is insulting you	55	45
If someone is annoying you	45	55

K. LEDs/BUTTONS

[illegible]

F. SWD/DEBUG

Circumstance	Justified (%)	Not justified (%)
If someone is attacking you	85	15
If someone is threatening you	75	25
If someone is harassing you	65	35
If someone is insulting you	55	45
If someone is annoying you	15	85

L. TEST POINTS

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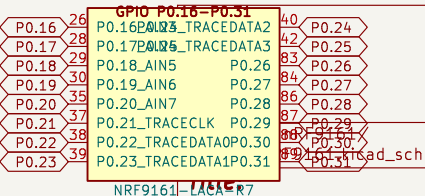
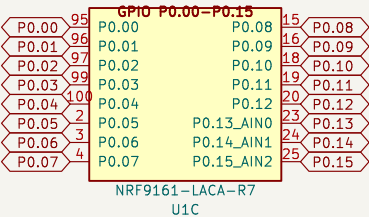
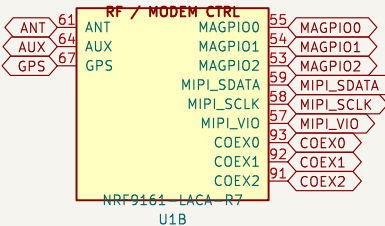
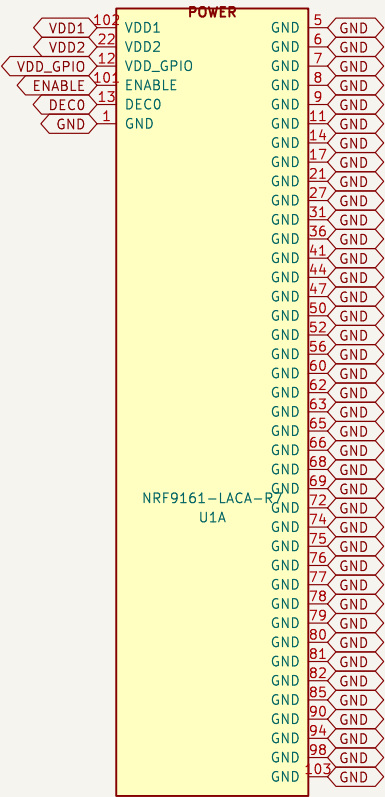
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Electrical source of truth: nRF9161 Product Specification v1.0 Table 55 + Figure 122
nRF91 Hardware Design Guidelines decoupling, nRF9161 DK 1.0.0 RF/GNSS/SIM reference.
Do not populate C18–C21 RF match until antenna/VNA characterization (Nordic OR pi default).

A. nRF9161-LACA-R7 (Nordic PS v1.0 Table 55)

SIM_DET (45) MUST FLOAT per PS. RES pins soldered mechanically, electrically NC. GND_Shield tied to VSS pin 103.

nRF9161 PIN POLICY (PS v1.0)
RES pins 10,51,70,71,73,104-127: solder for mechanical rigidity, electrically NC
SIM_DET pin 45: MUST FLOAT - do not wire (PS: Not used)
888 pins only. GND pin 45 must float (PS). DECO=4.7uF.
DECO pin 13: 4.7uF only (PCN134 / nRF91x1)



B. POWER (nRF9161 PS 11.3.1 + nRF91 HDG + DK FB3 on VDD2)



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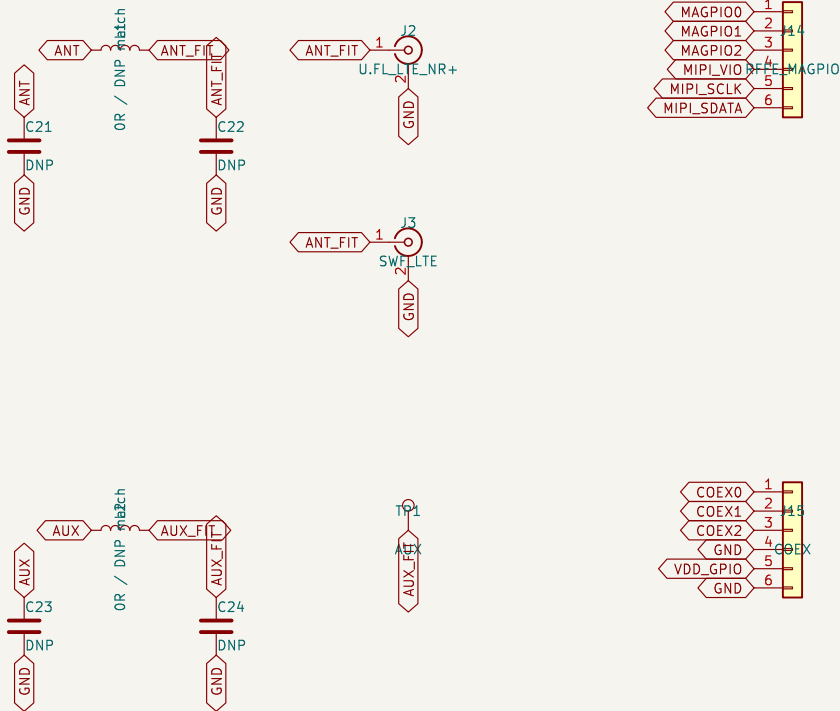
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The schematic diagram illustrates the electrical connections for a GNSS module. Key components and connections include:

- Capacitors:** C28 (100pF), C29 (10nF), C30 (100nF), C31 (DNP), and C32 (DNP) are connected to various signal and power lines.
- Inductor:** L4 (100nH) is connected between GNSS_VBIAS and GNSS_ANT.
- Resistor:** R4 (OR) is connected between COEX0 and GNSS_LNA_EN.
- Component:** BLM15H610ZB1 is connected to VDD_GPIO and GNSS_VBIAS.
- GNSS Signals:** GNSS_VBIAS, GNSS_ANT, GNSS_LNA, and GNSS_LNA_EN are shown as signal lines.
- Power and Ground:** Connections for GND, VDD_GPIO, and GPS are shown.
- Connectors:** J5 (U.FL GNSS) and J6 (SWF GNSS) are connected to GNSS_ANT.

The diagram is organized on a grid with columns 1-5 and rows A-D.

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The schematic diagram shows a nanoSIM_4FF component with the following connections and components:

- Capacitors:** C35, C36, C37, C38 (22pF) connected to SIM_CLK, SIM_RST, SIM_IO, and SIM_1V8 respectively to GND. C33 (100nF) connected to SIM_1V8 to GND. C34 (220nF) connected to SIM_VCC to GND.
- Resistor:** R4 connected between SIM_1V8 and SIM_VCC.
- Truth Table:**

J7	C1	VCC	C2	RST	C3	CLK	C5	GND	C6	VPP/SWP	C7	I/O	SIM_IO_C	DSW	SIM_CD	CSW	SH	GND	SH
- Other Components:** nanoSIM_4FF, PWR_VCC_LAG, and various signal pins (SIM_CLK, SIM_RST, SIM_IO, SIM_1V8, SIM_VCC, SIM_CD, SIM_CD, SIM_CD).

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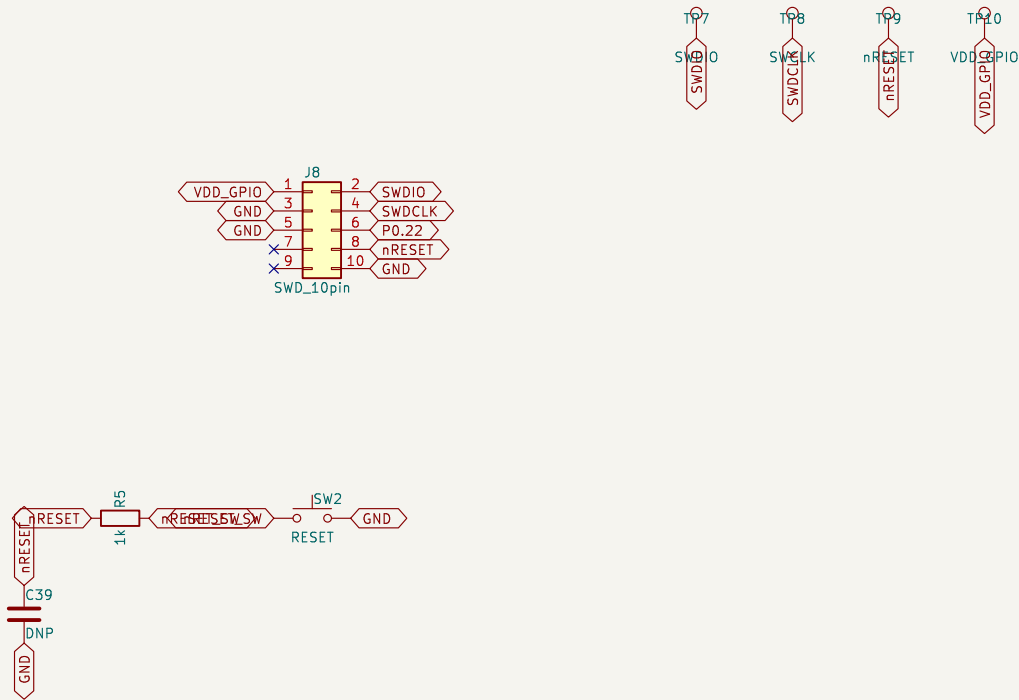
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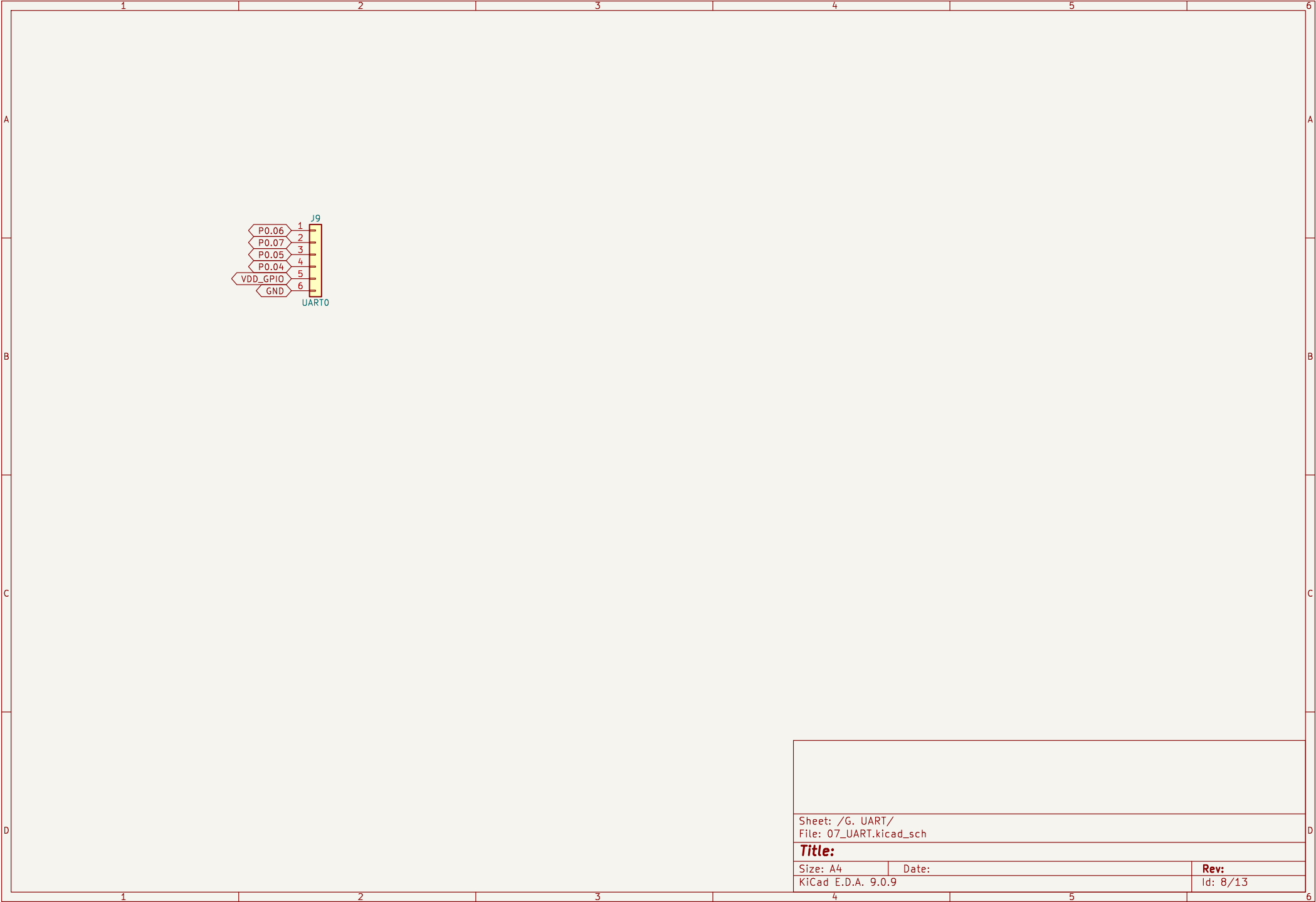
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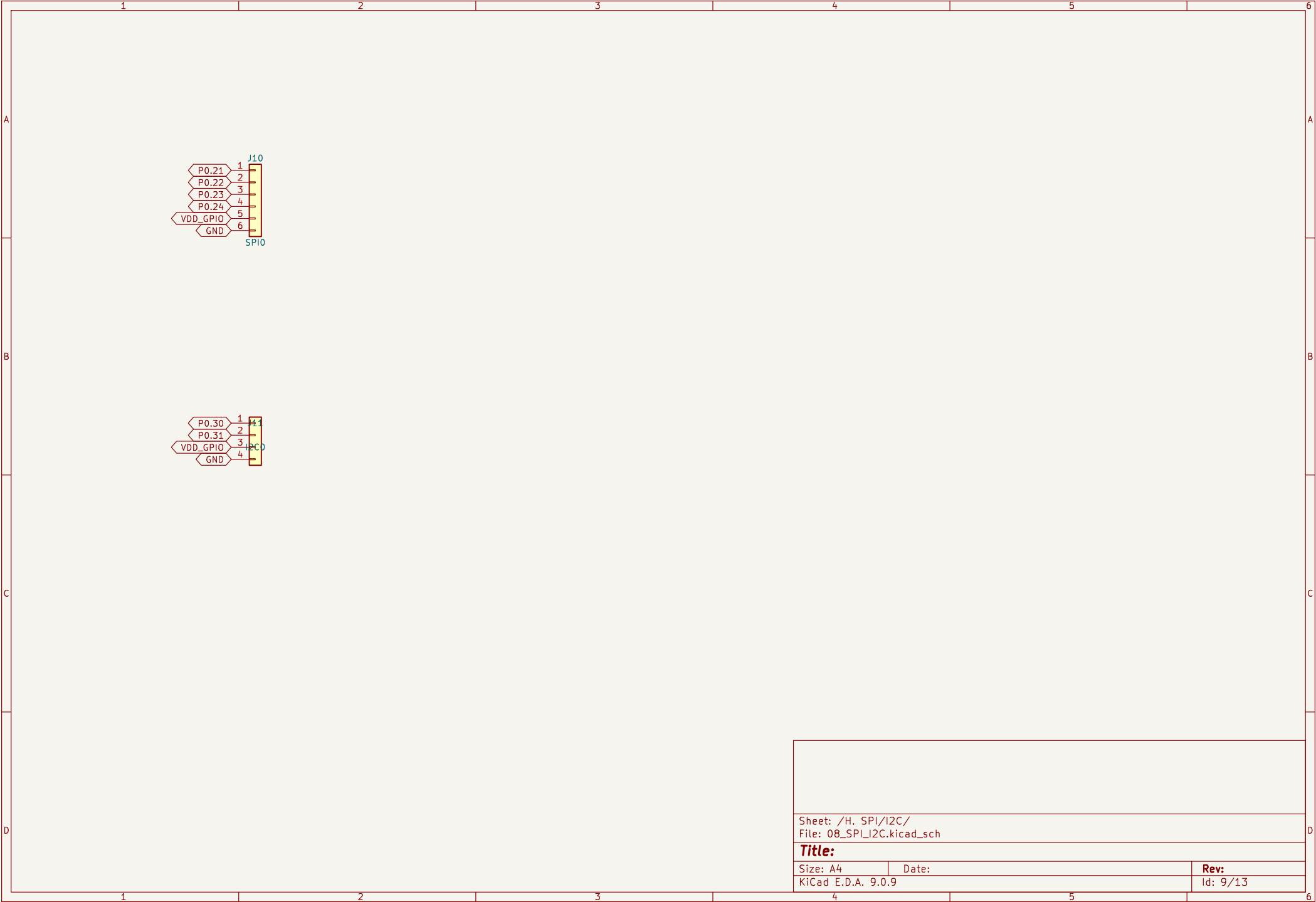
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F. SWD/DEBUG (10-pin Cortex-M, J-Link compatible)

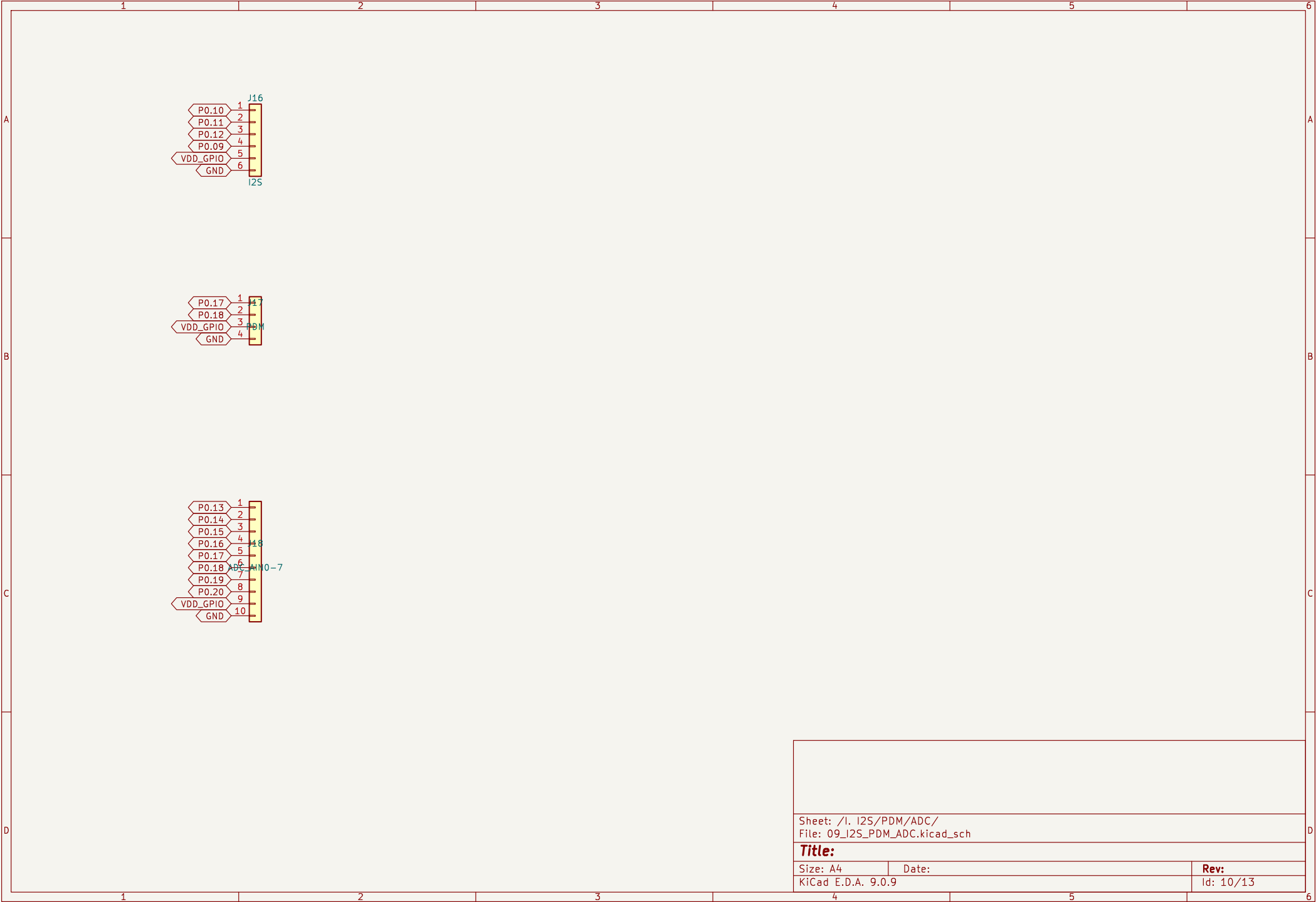




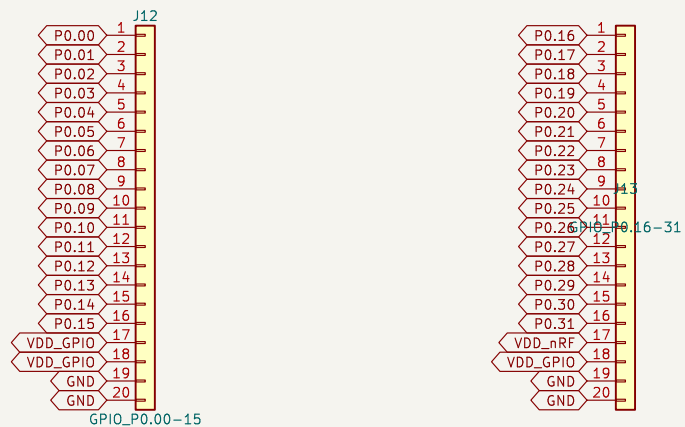
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P0.23 pin39 SPI_MISO (J10)
P0.20 pin16 I2S_SCK MAP (NRF9161 PS Table 55). All 32 GPIOs on J12/J13.
P0.28 P0.031 HPAE_BDRIVEN=drain OK; still on headers)
P0.24 UART_TXCS P0.05 UART_RTS P0.06 UART_TX P0.07 UART_RX (J9 default)
P0.20 button/button SW3, 10k to VDD_GPIO
P0.28 P0.031 I2S_SCL P0.10 I2S_SCK P0.11 I2S_LRCK P0.12 I2S_SDOUT (J16)
P0.25 pin20 CAN_AIN7 (J18) P0.17/P0.18 also PWM CLK/DIN (J17)
P0.24 pin3 SCL_HACECLK P0.22 SPI_MOSI / SDA P0.23 SPI_MISO P0.24 SPI_CS (J10)
Headers are labeled only except P0.28 expansion header labels only – GPIOs are not dedicated in hardware except LEDs/USER.



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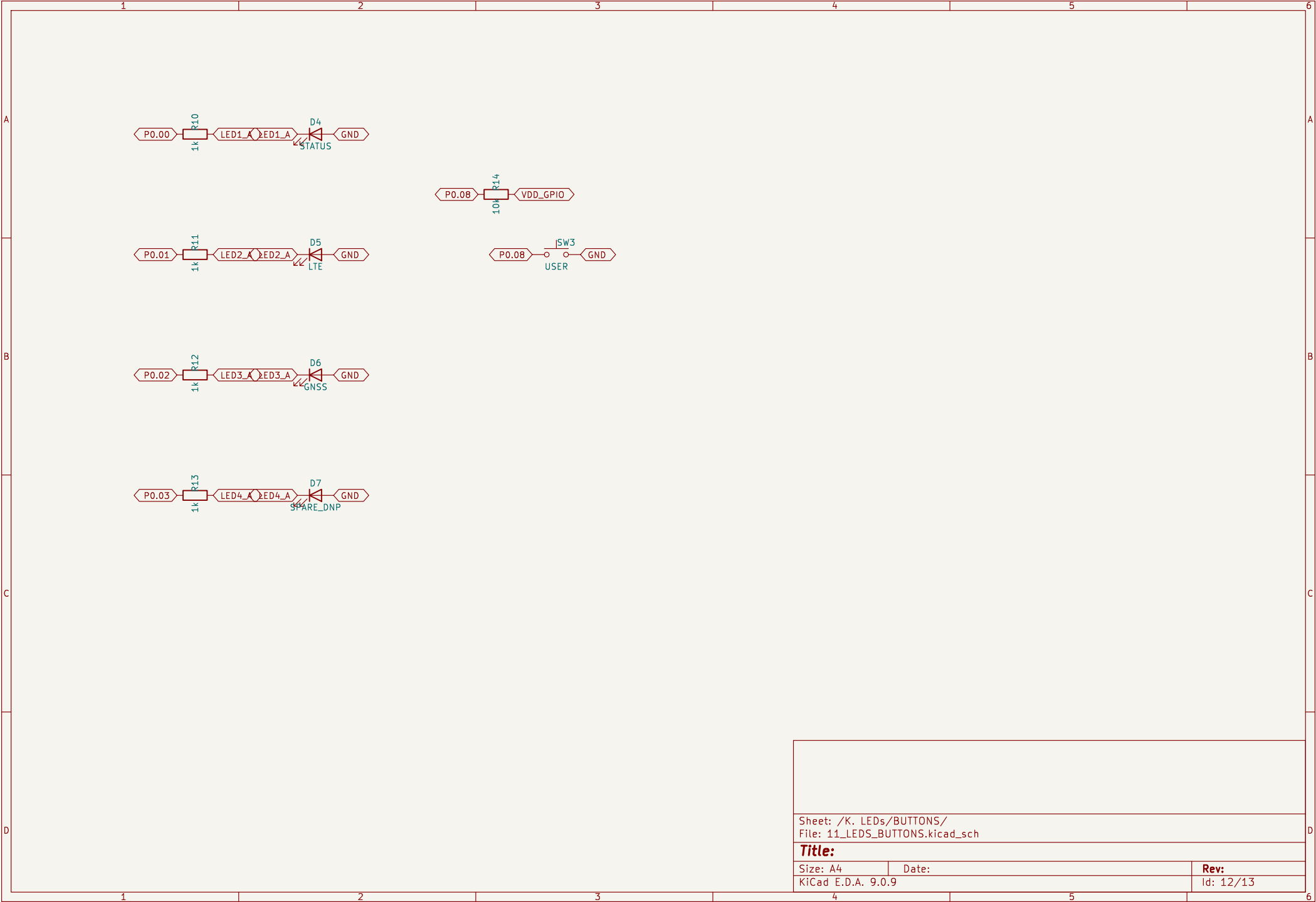
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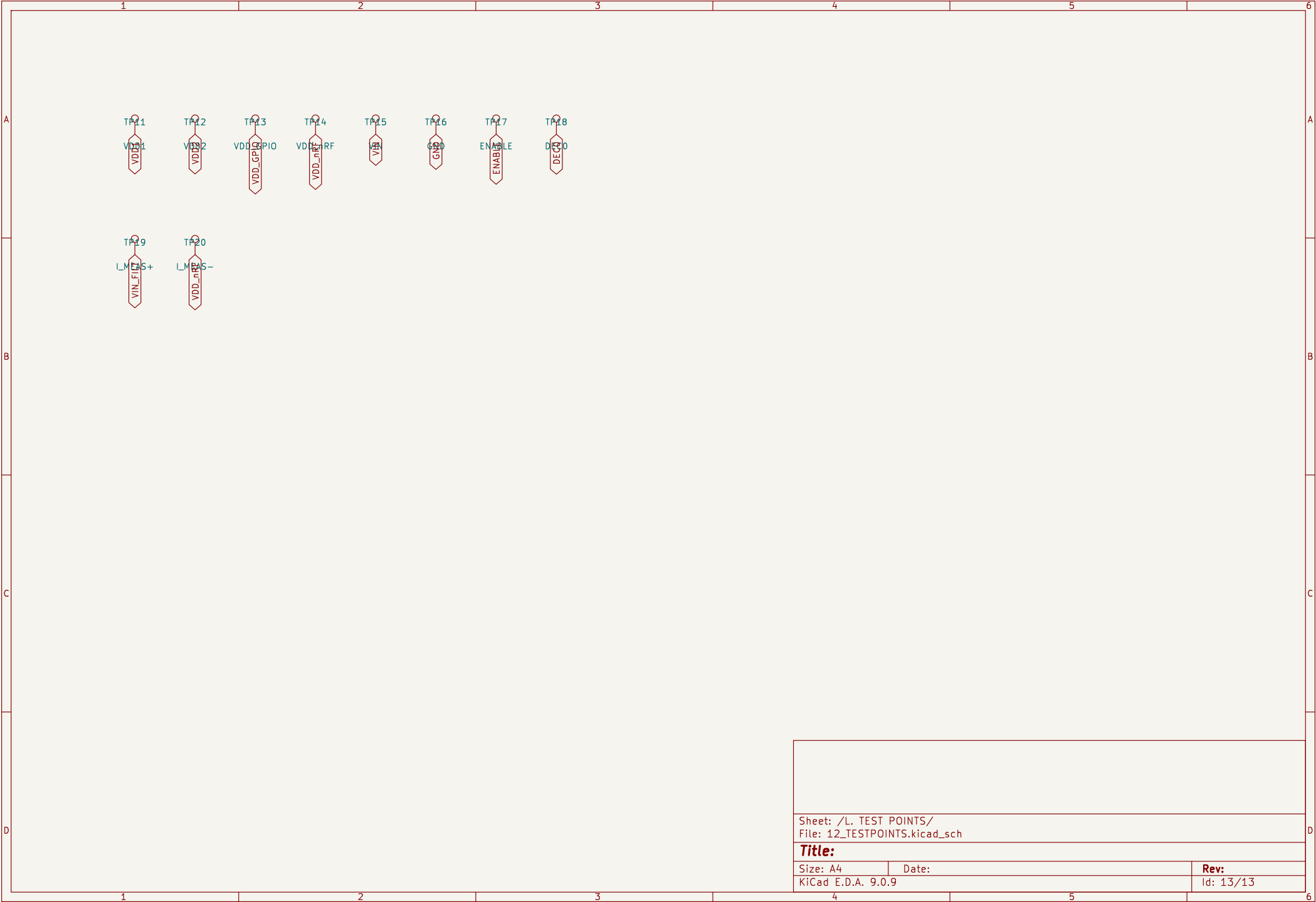
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